

bluetoothcar.c File Reference

```
#include <SoftwareSerial.h>
```

Functions

SoftwareSerial Bluetooth (BT_RX , BT_TX)
SoftwareSerial object for Bluetooth communication.
void setup ()
Arduino initialization.
void loop ()
Main execution loop.
void forward ()
Drive both motors forward.
void backward ()
Drive both motors backward.
void stop_the ()
Stop both motors completely.
void leftward ()
Turn left by slowing the left motor and running the right motor at full speed.
void rightward ()
Turn right by slowing the right motor and running the left motor at full speed.

Variables

const int BT_RX = 2
Bluetooth module RX pin (Arduino reads from this pin). Connect HC-05 → TX.
const int BT_TX = 3
Bluetooth module TX pin (Arduino writes to this pin). Connect HC-05 → RX (with voltage divider).
const int LEFT_IN1 = 9
Left motor input pin 1.
const int LEFT_IN2 = 10
Left motor input pin 2.
const int LEFT_EN = 5
Left motor enable pin (PWM)
const int RIGHT_IN1 = 11
Right motor input pin 1.
const int RIGHT_IN2 = 12
Right motor input pin 2.

```
const int RIGHT_EN = 6
```

Right motor enable pin (PWM)

```
int speed = 200
```

Default motor speed (0–255).

```
char t
```

Stores incoming Bluetooth command character.

Function Documentation

◆ backward()

```
void backward ( )
```

Drive both motors backward.

◆ Bluetooth()

```
SoftwareSerial Bluetooth ( BT_RX ,  
                           BT_TX )
```

SoftwareSerial object for Bluetooth communication.

◆ forward()

```
void forward ( )
```

Drive both motors forward.

◆ leftward()

```
void leftward ( )
```

Turn left by slowing the left motor and running the right motor at full speed.

◆ loop()

```
void loop ( )
```

Main execution loop.

Reads characters from Bluetooth and calls the corresponding motor control function.

◆ **rightward()**

```
void rightward ( )
```

Turn right by slowing the right motor and running the left motor at full speed.

◆ **setup()**

```
void setup ( )
```

Arduino initialization.

Sets pin modes and starts Bluetooth serial communication.

◆ **stop_the()**

```
void stop_the ( )
```

Stop both motors completely.

Variable Documentation

◆ **BT_RX**

```
const int BT_RX = 2
```

Bluetooth module RX pin (Arduino reads from this pin). Connect HC-05 → TX.

◆ **BT_TX**

```
const int BT_TX = 3
```

Bluetooth module TX pin (Arduino writes to this pin). Connect HC-05 → RX (with voltage divider).

◆ LEFT_EN

```
const int LEFT_EN = 5
```

Left motor enable pin (PWM)

◆ LEFT_IN1

```
const int LEFT_IN1 = 9
```

Left motor input pin 1.

◆ LEFT_IN2

```
const int LEFT_IN2 = 10
```

Left motor input pin 2.

◆ RIGHT_EN

```
const int RIGHT_EN = 6
```

Right motor enable pin (PWM)

◆ RIGHT_IN1

```
const int RIGHT_IN1 = 11
```

Right motor input pin 1.

◆ RIGHT_IN2

```
const int RIGHT_IN2 = 12
```

Right motor input pin 2.

◆ speed

```
int speed = 200
```

Default motor speed (0–255).

◆ t

```
char t
```

Stores incoming Bluetooth command character.